M Khinshan Arif

+92 303 0694747 | mkhinshanarif11@gmail.com | LinkedIn: Muhammad Khinshan Arif | Github: mkhinshanarif

WORK EXPERIENCE

Knowledge Streams | **Artificial Intelligence| Soft Skills Trainee** *(Jan 2026 – Apr 2026)*

- Applied **EDA, statistical analysis, and machine learning** using Python (Pandas, NumPy, Scikit-Learn, TensorFlow), improving data accuracy by **~25%** and generating actionable insights through visualization (Matplotlib, Seaborn).
- Collaborated with cross-functional teams to deliver ML-driven solutions, optimizing application performance with **~30% latency reduction** and **~20% increase in user satisfaction**.
- Demonstrated strong **analytical thinking, time management, teamwork, and communication skills** through end-to-end project execution and research-driven problem solving.

Technologies: (Python, Numpy, Pandas, Matplotlib, Seaborn, etc.)

PROJECTS

SignSpeak (Edge AI)

- Architected a high-performance **ASL translation dashboard** using MediaPipe (21-point 3D hand landmarks) and a custom **TensorFlow.js DNN**, enabling real-time gesture recognition.
- Optimized inference to **30+ FPS** by implementing **Wrist-Relative Normalization**, transforming raw coordinates into distance-invariant tensors and achieving **~98% accuracy**.
- Built a **privacy-first Edge AI solution** with browser-side processing (Keras → TF.js), integrating **real-time TTS** and dual-mode input (webcam + image) for enhanced accessibility.

Technologies: (JavaScript, TensorFlow.js, MediaPipe, Python)

Virtual Mouse

- Developed a **Virtual Mouse application** using Python, OpenCV, and MediaPipe, achieving **95% gesture recognition accuracy** and enabling touch-free, real-time cursor control.
- Collaborated with team members to optimize hand-tracking and gesture-mapping algorithms using NumPy and PyAutoGUI, **reducing development time by 20%**, enhancing system responsiveness by 30%, and incorporating user feedback to deliver a **seamless and intuitive interaction experience**.

Technologies:(Python, OpenCV, MediaPipe, NumPy, PyAutoGUI, Math Library)

Event-Desk-Flask Website:

- Provides a secure platform where users can **register, view, and add events**, see **upcoming events**, generate QR codes, and make online payments—simplifying event and account management**.**
- Designed for individuals or businesses managing events who need QR code generation and payment functionality; my role is **backend development**, handling routes, database, authentication, event logic, and Stripe integration.
- Backend with **Python Flask**, database via **SQL Alchemy** and **Flask-Migrate**, frontend with **Flask-Bootstrap**, forms with **Flask-WTF**, QR code generation with **qrcode**, and payment integration using **Stripe**.

CORE SKILLS

Programming Languages: Python, C++
Technical Skills: Python, Numpy, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn, TensorFlow, Keras, EDA.
Developer Tools: Jupyter, VS Code, SQL, Google Colab
Frameworks: Flask
Databases: SQL, MySQL, SQL Alchemy

CERTIFICATIONS

Crash Course in Data Science (Coursera) (July, 2025)
Data Analysis with Python and Pandas (LinkedIn) (March, 2026)
Microsoft Azure AI Essentials (April, 2026)

EDUCATION

The Superior University Lahore
BS Computer Science (2023-Present)